Introductory Mathematical Economics

Continuity

Delhi School of Economics

Continuity

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Reading: Simon and Blume Ch. 14

Function of Many Variables

[We shall restrict our attention to functions from $\mathcal{R}^l$ to $\mathcal{R}$.]

Main question: Is there an abrupt change in the output (functional value)?

Informal idea: A function is 'smooth' at an input $c \in \mathcal{R}^l$ if inputs which are 'indistinguishable' from c generate output which is 'indistinguishable' from $f(c)$.

Web-link

Surface Plot:
https://academo.org/demos/3d-surface-plotter/
Contour Plot:
https://academo.org/demos/contour-plot/

Example-discontinuous function

$$f(x,y) = \begin{cases} -0.5 & \text{if } -0.2 \leq x \leq 0.2 \text{ and } -0.2 \leq y \leq 0.2 \\ x^2 - y^2 & \text{otherwise} \end{cases}$$

This function is discontinuous at all points on the edges of the rectangle $S = \{(x,y) | -0.2 \leq x \leq 0.2 \text{ and } -0.2 \leq y \leq 0.2\}$

Continuity: Definition I

f is **continuous** at $c \in \mathcal{R}^l$ if for all sequence $\{x_n\}_{n=1}^{\infty}$ converging to c, we have $\{f(x_n)\}_{n=1}^{\infty}$ converging to $f(c)$

Continuity Definition II

f is **continuous** at $c \in \mathcal{R}^l$, if for every $\epsilon > 0$, there exists $\delta > 0$ such that $x \in B_\delta(c)$ implies $f(x) \in B_\epsilon(f(c))$.

Continuity

If f is continuous at every point in the domain, then f is called **continuous**.

Result: Two definitions of continuity are equivalent.

[A proof is available at the end. It us recommended but optional.]

Examples

- $f : \mathcal{R} \to \mathcal{R}$, $f(x) = |x|$ is continuous at 0. In fact f is continuous at all points.
- $f : \mathcal{R} \to \mathcal{R}$, $f(x) = \lceil x \rceil$ = smallest integer $\geq x$. f is not continuous at integers.
- $f : \mathcal{R} \to \mathcal{R}$,

$$f(x) = \begin{cases} \sin\left(\frac{1}{x}\right) & \text{if } x \neq 0 \\ 0 & \text{if } x = 0 \end{cases}$$

  f is discontinuous at 0
- $f : \mathcal{R}^2 \to \mathcal{R}$, $f(x, y) = xy$ is a continuous function.
- $f : \mathcal{R}^2 \to \mathcal{R}$,

$$f(x, y) = \begin{cases} \frac{xy}{x^2 + y^2} & \text{if } (x, y) \neq (0, 0) \\ 0 & \text{if } (x, y) = (0, 0) \end{cases}$$

  f is discontinuous at $(0, 0)$

Useful Results

Let $f : A \to \mathcal{R}$ and $g : A \to \mathcal{R}$ are continuous at $c \in A$. Then

- For $\alpha \in \mathcal{R}$, $\alpha f(x)$ is continuous at c.
- $f(x) + g(x)$ is continuous at c.
- $f(x)g(x)$ is continuous at c.
- If, $g(c) \neq 0$ then $\frac{f(x)}{g(x)}$ is continuous at c.
- $f : A \to \mathcal{R}$ is continuous at $c \in A$. $h : \mathcal{R} \to \mathcal{R}$ is continuous at $f(c)$. Then $f \circ h$ is continuous at c.

Intermediate Value Theorem

Intermediate Value Theorem: $f : \mathcal{R} \to \mathcal{R}$ is continuous function such that $f(a) > 0$ and $f(b) < 0$ (or the opposite). Then there exists $c \in [a, b]$ such that $f(c) = 0$.

Result $f : \mathcal{R} \to \mathcal{R}$ is continuous function at c. If $f(c) > 0$ then there exists $\delta > 0$ such that for all $x \in (c - \delta, c + \delta)$, $f(x) > 0$.

Intermediate Value Theorem - Proof

Result $f : \mathcal{R} \to \mathcal{R}$ is continuous function at c. If $f(c) > 0$ then there exists $\delta > 0$ such that for all $x \in (c - \delta, c + \delta)$, $f(x) > 0$.

Proof: Since f is continuous at c, we can find a δ such that for all $x \in (c - \delta, c + \delta)$, we have $|f(x) - f(c)| < \frac{f(c)}{2}$.
That is for all $x \in (c - \delta, c + \delta)$, we have $0 < \frac{f(c)}{2} < f(x) < \frac{3f(c)}{2}$. Hence proved.

Similarly, if f is continuous function at c and $f(c) < 0$ then there exists $\delta > 0$ such that for all $x \in (c - \delta, c + \delta)$, $f(x) < 0$.

Intermediate Value Theorem: $f : \mathcal{R} \to \mathcal{R}$ is continuous function such that $f(a) > 0$ and $f(b) < 0$ (or the opposite). Then there exists $c \in [a, b]$ such that $f(c) = 0$.

Proof: Lets take the case $f(a) > 0$ and $f(b) < 0$. Define $S = \{x \in [a, b] \mid f(x) > 0\}$.
S is non-empty because $a \in S$. S is bounded by construction, Thus S has a supremum, denote it by c. c must be smaller than b because $b \notin S$. We want to show that $f(c) = 0$.
Suppose that $f(c) > 0$. By Lemma, there is an interval $(c, c + \delta_1)$ such that for all $x \in (c, c + \delta_1)$, $f(x) > 0$. This contradicts the fact that c is supremum of S.
Now suppose that $f(c) < 0$. By Lemma, there is an interval $(c - \delta_2, c)$ such that for all $x \in (c - \delta_2, c)$, $f(x) < 0$. Then c can not be the supremum (second definition of supremum).
Hence $f(c) = 0$.

Equivalence of two definitions - Proof (Optional)

Result Two definitions of continuity are equivalent.

Proof: First defn. $\Rightarrow$ Second defn.: Let $\lim_{x\to c} f(x) = f(c)$ by first definition. We shall prove by contradiction. Suppose there exists an ϵ, for which we can not find a δ. That is for for all δ, there exist a point x, which is in δ-neighbourhood of c but $f(x)$ is not in ϵ-neighbourhood of $f(c)$.
We can choose $\delta = \frac{1}{k}$ for $k = 1, 2, \ldots$. For each we shall get x_k such that $|x_k - c| < \frac{1}{k}$ but $|f(x_k) - f(c)| \geq \epsilon$.
Note that $\{x_k\}_{k=1}^{\infty}$ is a sequence which converges to c.
However $\{f(x_k)\}_{k=1}^{\infty}$ does not converge to $f(c)$, because the distance between $f(x_k)$ and $f(c)$ is greater than ϵ for all k.
We have reached a contradiction.

Second defn. $\Rightarrow$ First defn.: Let $\lim_{x\to c} f(x) = f(c)$ by second definition. Take a sequence $\{x_k\}_{k=1}^{\infty}$ which converges to c. We want to show that $\{f(x_k)\}_{k=1}^{\infty}$ converges to $f(c)$.
Take any $\epsilon > 0$. By second definition, we can find a $\delta > 0$ such that for all x in $|x - c| < \delta$ implies $|f(x) - f(c)| < \epsilon$.
Since $\lim x_k = c$, given δ, I can find N such that $|x_k - c| < \delta$ for all $k \geq N$. Hence for all $k \geq N$, $|f(x_k) - f(c)| < \epsilon$. Therefore $\{f(x_k)\}_{k=1}^{\infty}$ converges to $f(c)$.